

VPMP POLYTECHNIC, GANDHINAGAR

APPLIED MATHEMATICS

COURSE CODE: - DI02000011



ASSIGNMENT



UNIT-I

MARKS-16

MATRICES



MATRICES

➤ MCQ'S

- 1) Order of $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$ is ____.
- 2) Order of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix}$ is ____.
- 3) Order of $\begin{bmatrix} 2 & -1 \\ 5 & 3 \end{bmatrix}$ is ____.
- 4) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then $A^T =$ ____.
- 5) If $A = \begin{bmatrix} 1 & -3 & 4 \\ -2 & 1 & 2 \end{bmatrix}$ then $A^T =$ ____.
- 6) If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then $A^T =$ ____.
- 7) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 4 & 2 \end{bmatrix}$ then $A^T =$ ____.
- 8) If A is non-singular matrix then ____.
- 9) If $A = \begin{bmatrix} 1 & 4 \\ 3 & -2 \end{bmatrix}$ then $3A =$ ____.
- 10) If $\begin{bmatrix} 3 & 2 \\ x-1 & 5 \end{bmatrix} = \begin{bmatrix} 3 & y+1 \\ 4 & 5 \end{bmatrix}$ then $(x, y) =$ ____.
- 11) Find x and y if $\begin{bmatrix} x+y & 3 \\ -7 & x-y \end{bmatrix} = \begin{bmatrix} 8 & 3 \\ -7 & 2 \end{bmatrix}$.
- 12) $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} =$ ____.
- 13) If $A = \begin{bmatrix} 2 & -1 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ then $2A + 3B =$ ____.
- 14) If $A = \begin{bmatrix} 1 & 4 \\ 3 & -2 \end{bmatrix}$ then $2A - 3I =$ ____.
- 15) If $A_{3 \times 4}$ and $B_{4 \times 4}$ are the matrices then the order of AB is ____.
- 16) If $A_{2 \times 3}$ and $B_{3 \times 4}$ are the matrices then the order of $(AB)^T$ is ____.
- 17) If $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ then $AB =$ ____.
- 18) If $A = [1 \ 2 \ 3]$ and $B = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ then $AB =$ ____.
- 19) If $A = [2 \ 0 \ 5]$ and $B = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ then $BA =$ ____.
- 20) If $A = \begin{bmatrix} -7 & 6 \\ 5 & -2 \end{bmatrix}$ then $AI =$ ____.
- 21) If $[0 \ x \ -2] \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = [4]$ then $x =$ ____.
- 22) If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then $A^2 =$ ____.
- 23) If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $adjA =$ ____.



- 24) If $A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$ then $\text{adj}A = \underline{\hspace{2cm}}$.
- 25) If $A = \begin{bmatrix} -8 & 4 \\ -6 & 3 \end{bmatrix}$ then $A^{-1} = \underline{\hspace{2cm}}$.
- 26) If $A = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}$ then $A^{-1} = \underline{\hspace{2cm}}$.
- 27) If $AB = I$ then matrix B = $\underline{\hspace{2cm}}$.



➤ **EXAMPLES (Based on Addition , subtraction and equality)**

- 1) If $A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & 5 \\ -1 & 2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 0 & 7 \\ 2 & 3 & -4 \end{bmatrix}$ then find $2A + 3B$.
- 2) If $A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ then find $3A - 2B$.
- 3) If $A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & 0 & 2 \\ 4 & 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 & 5 \\ 5 & 4 & 3 \\ 3 & 5 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 2 \\ 4 & 5 & 6 \end{bmatrix}$ then find $3A + 2B - 4C$.
- 4) If $A = \begin{bmatrix} 2 & 3 & 6 \\ -1 & 2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 & -8 \\ 2 & 4 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 3 & -3 \\ 1 & 4 & 1 \end{bmatrix}$ then prove that $2A + 3B - 4C = 0$.
- 5) If $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -2 & 4 \\ 1 & 5 & 0 \end{bmatrix}$ then find matrix X from $X + A + B = 0$.
- 6) If $A = \begin{bmatrix} 1 & 2 & 0 \\ -3 & 2 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 & 3 \\ -2 & 3 & 2 \end{bmatrix}$ then find matrix X from $3(X + B) + 5A = 0$.
- 7) If $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix}$ then prove that $(A + B)^T = A^T + B^T$.
- 8) If $M = \begin{bmatrix} -2 & 3 & 8 \\ 5 & -7 & 9 \\ 1 & -4 & 6 \end{bmatrix}$, $N = \begin{bmatrix} 15 & -6 & 2 \\ 11 & 4 & 7 \\ 13 & 5 & 6 \end{bmatrix}$ then prove that $(M + N)^T = M^T + N^T$.
- 9) If $3A - 2B = \begin{bmatrix} 2 & 1 \\ -2 & -3 \end{bmatrix}$ and $B - 4A = \begin{bmatrix} -1 & 2 \\ -4 & 4 \end{bmatrix}$ then find A and B .
- 10) If $\begin{bmatrix} a + 2b & 3c + 2d \\ 2a - b & c - d \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 3 & 5 \end{bmatrix}$ then find the values of a, b, c and d .

➤ **EXAMPLES (Based on multiplication)**

- 1) If $A = \begin{bmatrix} -1 & 2 & 3 \\ 3 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 5 \end{bmatrix}$ find AB or BA whichever exist.
- 2) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}$ then find AB and BA .
- 3) If $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ then show that $A^2 = A$.
- 4) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then prove that $A^2 - 5A - 2I = 0$.
- 5) If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ then prove that $A^2 - 4A + 7I = 0$.
- 6) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then prove that $A^2 - 4A - 5I = 0$.
- 7) If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ then prove that $A^4 = I$.
- 8) If $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 2 & 1 \end{bmatrix}$ then find $(AB)^T$.
- 9) If $A = \begin{bmatrix} 2 & -2 \\ 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 \\ 4 & -3 \end{bmatrix}$ then prove that $(AB)^T = B^T \cdot A^T$.



10) From equation $\begin{bmatrix} x & 3 \\ y & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \end{bmatrix}$ find value of x and y .

11) Evaluate $\begin{bmatrix} 2 & 1 & -1 \\ 4 & -5 & 6 \\ -3 & 7 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -6 & 4 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -2 & 1 \end{bmatrix}$.

➤ **EXAMPLES (Based on adjoint and inverse)**

1) If $A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ then prove that $\text{adj}A = A$.

2) If $A = \begin{bmatrix} 5 & -3 \\ 2 & 1 \end{bmatrix}$ then find A^{-1} .

3) If $A = \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$ then find A^{-1} .

4) Find matrix X such that $\begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix} \cdot X = \begin{bmatrix} -16 & -6 \\ 7 & 2 \end{bmatrix}$.

5) If $A + B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}$ and $A - B = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$ then find $(AB)^{-1}$.

6) Find A^{-1} if exists for $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$.

7) Find inverse of matrix $\begin{bmatrix} 3 & -1 & 2 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{bmatrix}$.

8) If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ then find A^{-1} .

9) If $A = \begin{bmatrix} 3 & -10 & -1 \\ -2 & 8 & 2 \\ 2 & -4 & -2 \end{bmatrix}$ then find A^{-1} .

10) If $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ then find A^{-1} .

➤ **EXAMPLES (Based on simultaneous linear equation)**

1) Solve the equation $3x - y = 1$ and $x + 2y = 5$ using matrix method.

2) Solve the equation $2x + 5y = 7$ and $8x - 3y = 5$ using matrix method.

3) Solve the equation $2x + 3y = 1$ and $y - 4x = 2$ using matrix method.

4) Solve the equation $2x - 3y = -5$ and $3x + y = 9$ using matrix method.

5) Solve the equation $3x + 2y = 7$ and $11x - 4y = 3$ using matrix method.

6) Solve the equation $3x + 2y = 5$ and $2x - y = 1$ using matrix method.

7) Solve the equation $2x + 3y = 6xy$ and $x - y = xy$ using matrix method.



UNIT-II

MARKS-16

DIFFERENTIATION

AND

ITS APPLICATION



DIFFERENTIATION

➤ MCQ'S

- 1) $\frac{d}{dx}(\sin^2 x + \cos^2 x) = \underline{\hspace{2cm}}$.
- 2) If $f(x) = 2^x$ then $f'(0) = \underline{\hspace{2cm}}$.
- 3) $\frac{d}{dx}(x \log x) = \underline{\hspace{2cm}}$.
- 4) If $f(x) = e^{2x}$ then $f'(x) = \underline{\hspace{2cm}}$.
- 5) $\frac{d}{dx} \log(\cos x) = \underline{\hspace{2cm}}$.
- 6) $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = \underline{\hspace{2cm}}$.
- 7) $\frac{d}{dx}(x^x) = \underline{\hspace{2cm}}$.
- 8) $x^2 + y^2 = 29$ then $\frac{dy}{dx}$ at the point (2,5) = $\underline{\hspace{2cm}}$.
- 9) $\frac{d}{dx}(\cot x) = \underline{\hspace{2cm}}$.
- 10) $\frac{d(\log \sin x)}{dx} = \underline{\hspace{2cm}}$.
- 11) $\frac{d(\frac{u}{v})}{dx} = \underline{\hspace{2cm}}$.
- 12) $\frac{d(e^{-\log x})}{dx} = \underline{\hspace{2cm}}$.
- 13) If $x = \sec \theta + \tan \theta$ and $y = \sec \theta - \tan \theta$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- 14) If $f(x) = \log \sqrt{x^2 + 1}$ then $f'(0) = \underline{\hspace{2cm}}$.
- 15) If $x = at$ and $y = \frac{a}{t}$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- 16) If $\frac{d}{dx}(x^n) = nx^{n-1}$ then $\frac{d}{dx}(y^4) = \underline{\hspace{2cm}}$.
- 17) $\frac{d}{dx}(\sec x) = \underline{\hspace{2cm}}$.
- 18) If $x = \cos \theta$ and $y = \sin \theta$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- 19) If $x^2 + y^2 = 1$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- 20) If $y = \sin^{99} \left(\frac{\pi}{2} \right)$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- 21) If $y = \cot x$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- 22) If $y = e^x$ then $\frac{d^2 y}{dx^2} = \underline{\hspace{2cm}}$.
- 23) $\frac{d}{dx}(x^2 + 2^x + 2^2) = \underline{\hspace{2cm}}$.
- 24) $\frac{d}{dx}(\sqrt{x}) = \underline{\hspace{2cm}}$.
- 25) If $f(x) = \sin x$ then $f' \left(\frac{\pi}{2} \right) = \underline{\hspace{2cm}}$.
- 26) $\frac{d}{dx}(3 \sin x - 4 \sin^3 x) = \underline{\hspace{2cm}}$.
- 27) $\frac{d}{dx}(a^x) = \underline{\hspace{2cm}}$.
- 28) $\frac{d}{dx}(\tan^{-1} x + \cot^{-1} x) = \underline{\hspace{2cm}}$.



29) $\frac{d}{dx}(x^2 + 2x + 7) = \underline{\hspace{2cm}}$.

30) $\frac{d}{dx}(\tan x) = \underline{\hspace{2cm}}$.

31) $\frac{d}{dx}(\tan^2 x - \sec^2 x) = \underline{\hspace{2cm}}$.

32) If $y = 1,00,000$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

33) If $y = \tan x$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

34) If $y = e^x + 4$ then $\frac{d^2y}{dx^2} = \underline{\hspace{2cm}}$.

35) $\frac{d}{dx}(\operatorname{cosec} x) = \underline{\hspace{2cm}}$.

36) $\frac{d}{dx}(x^n) = \underline{\hspace{2cm}}$.



➤ **EXAMPLES (Based on Definition)**

- 1) Using definition of derivative, find the derivative of x^2 with respect to x .
- 2) Find the derivative of x^3 using definition.
- 3) Using first principle of differentiation, differentiate $\sqrt{x}$ with respect to x .
- 4) Differentiate a^x using definition.
- 5) Find derivative of e^x using definition.
- 6) Find derivative of $\sin x$ using definition.
- 7) Differentiate $f(x) = \cos x$ with respect to x by using definition of derivative.
- 8) Differentiate $x^2 + 2x - 1$ with respect to x by using first principle.
- 9) Differentiate $y = 2x^2 + 3$ using definition.

➤ **EXAMPLES (Based on Addition , subtraction , Multiplication , division ,Chain rule)**

- 1) Find $\frac{d}{dx}(x^3 \cdot \log x)$.
- 2) If $y = e^x \cdot \sin x$ then find $\frac{dy}{dx}$.
- 3) If $y = x^2 \cdot \tan x$ then find $\frac{dy}{dx}$.
- 4) Find $\frac{dy}{dx}$ when (I) $y = e^x \cdot \sec x$ (II) $y = \frac{\log x}{x}$.
- 5) For $y = \frac{\sin x}{x}$ find $\frac{dy}{dx}$.
- 6) Find the derivative of $\frac{\log x}{x}$ at $x=1$.
- 7) If $y = \frac{x^2-1}{x^2+1}$ then find $\frac{dy}{dx}$.
- 8) If $y = \frac{1+\sin x}{1-\sin x}$ then find $\frac{dy}{dx}$.
- 9) Find $\frac{dy}{dx}$ if $y = \frac{a+b\sin x}{b+a\sin x}$.
- 10) Find $\frac{dy}{dx}$ if $y = \frac{3+4\sin x}{4+3\sin x}$.
- 11) If $y = \frac{1+\tan x}{1-\tan x}$ then find $\frac{dy}{dx}$.
- 12) If $y = \log(\tan x) + \cos x + x$ then find $\frac{dy}{dx}$.
- 13) Find $\frac{dy}{dx}$ for $y = \log(\cos 2x)$.
- 14) If $y = x^3 \cdot \tan x + (\tan x)^3$ then find $\frac{dy}{dx}$.
- 15) Find $\frac{dy}{dx}$ if $y = \log(\operatorname{cosec} x - \cot x)$.
- 16) Find $\frac{dy}{dx}$ if $y = \log(\sec x + \tan x)$.
- 17) If $y = e^{3x} \cdot \cos 2x$ then find $\frac{dy}{dx}$.
- 18) If $y = e^{\sin x} \cdot \sec 2x$ then find $\frac{dy}{dx}$.
- 19) If $y = x^3 \cdot \sin(\log x)$ then find $\frac{dy}{dx}$.
- 20) If $y = \frac{\sin(\log x)}{x}$ then find $\frac{dy}{dx}$.
- 21) Find $\frac{dy}{dx}$ from $y = \log\left(\frac{\sin x}{1+\cos x}\right)$.
- 22) If $y = \log(x + \sqrt{x^2 + 1})$ then find $\frac{dy}{dx}$.



23) If $y = \log \sqrt{\frac{a+x}{a-x}}$ then find $\frac{dy}{dx}$.

➤ **EXAMPLES (Based on taking log on both side)**

- 1) If $y = x^x$ then find $\frac{dy}{dx}$.
- 2) If $y = x^{\sin x}$ then find $\frac{dy}{dx}$.
- 3) If $y = \sin x^x$ then find $\frac{dy}{dx}$.
- 4) If $y = (\sin x)^{\tan x}$ then find $\frac{dy}{dx}$.
- 5) If $y = (\sin x)^{\cos x}$ then find $\frac{dy}{dx}$.
- 6) If $y = x^x + (\sin x)^x$ then find $\frac{dy}{dx}$.

➤ **EXAMPLES (Based on Implicit function)**

- 1) If $x^2 + y^2 = xy$ then find $\frac{dy}{dx}$.
- 2) If $x^3 + y^3 = 3xy$ then find $\frac{dy}{dx}$.
- 3) If $x^3 + y^3 = 3axy$ then find $\frac{dy}{dx}$.
- 4) If $x + y = \sin(xy)$ then find $\frac{dy}{dx}$.
- 5) If $y = \sin(x + y)$ then find $\frac{dy}{dx}$.
- 6) If $y = \cos(x + y)$ then find $\frac{dy}{dx}$.
- 7) If $xsiny + ysinx = 5$ then find $\frac{dy}{dx}$.
- 8) If $ax^2 + 2hxy + by^2 = 0$ then find $\frac{dy}{dx}$.

➤ **EXAMPLES (Based on Parametric function)**

- 1) If $x = at^2$ and $y = 2at$ then find $\frac{dy}{dx}$.
- 2) If $x = a\sin\theta$ and $y = a(1 + \cos\theta)$ then find $\frac{dy}{dx}$.
- 3) If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$ then find $\frac{dy}{dx}$.
- 4) If $x = a(1 + \cos\theta)$ and $y = b(\theta + \sin\theta)$ then find $\frac{dy}{dx}$.
- 5) If $x = \sec\theta + \tan\theta$ and $y = \sec\theta - \tan\theta$ then find $\frac{dy}{dx}$.
- 6) If $x = \frac{1}{2}\left(t + \frac{1}{t}\right)$ and $y = \frac{1}{2}\left(t - \frac{1}{t}\right)$ then find $\frac{dy}{dx}$.
- 7) If $x = e^{2t}\cos t$ and $y = e^{2t}\sin t$ then find $\frac{dy}{dx}$.



➤ **EXAMPLES (Based on Successive differentiation)**

- 1) If $y = \log(\sin x)$ then prove that $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 1 = 0$.
- 2) If $y = \log(e^{\sin x})$ then prove that $\frac{dy}{dx} - \cos x = 0$.
- 3) If $y = e^{2x}$ then prove that $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.
- 4) If $x = at^2$ and $y = 2at$ then prove that $yy_2 + y_1^2 = 0$.
- 5) If $y = 2e^{3x} + 3e^{-2x}$ then prove that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$.
- 6) If $y = A \cos pt + B \sin pt$ then prove that $\frac{d^2y}{dt^2} + p^2y = 0$.

➤ **EXAMPLES (Based on application of differentiation Maximum and Minimum)**

- 1) Find maximum and minimum values of $f(x) = x^3 - 3x + 11$.
- 2) Find maximum and minimum values of $f(x) = 2x^3 - 3x^2 - 12x + 5$.
- 3) Find maximum and minimum values of $f(x) = 2x^3 - 15x^2 + 36x + 10$.
- 4) Find maximum and minimum values of $f(x) = x^3 + x^2 - x$.
- 5) Find maximum and minimum values of $f(x) = 3x^3 - 4x^2 - x + 5$.
- 6) Find maxima and minima of $f(x) = x + \frac{1}{x}$.

➤ **EXAMPLES (Based on application of differentiation Velocity and Acceleration)**

- 1) The equation of motion of particle is $s = t^3 - 6t^2 + 8t - 4$ then find velocity and acceleration of the moving particle at $t = 3$ second.
- 2) The distance of a moving particle is given by $s = t^3 - 3t^2 + 4t + 3$. Find velocity and acceleration at $t = 2$ second.
- 3) If the law of motion is $s = t^3 + 6t^2 + 3t + 5$ then find velocity and acceleration at $t = 3$ second.
- 4) The equation of motion of particle is $s = t^3 - 6t^2 + 9t$ then find velocity and acceleration at $t = 3$ second.
- 5) The motion of particle is given by $s = t^3 - 6t^2 + 9t + 6$ then find velocity when acceleration is zero.
- 6) The equation of motion of particle is $s = t^3 - 5t^2 + 3t$. When particle comes to rest? Find acceleration at that time.
- 7) A particle has motion of $s = 2t^3 + 3t^2 - 12t + 6$. Find velocity and acceleration at $t = 2$ second. Also find acceleration when particle comes to rest.
- 8) The equation of motion of particle is $s = t^3 - 6t^2 + 9t$ then find t and s at $v = 0$.
- 9) The equation of motion of particle is $s = t^3 - 6t^2 + 9t + 6$ where s is in meter and t is in second. (I) Find v and a when $t = 2$ second. (II) Find s when particle change its direction.
- 10) The equation of motion of particle is $s = t^3 + 3t$ ($t > 0$).
(I) Find velocity and acceleration at $t = 3$ second.
(II) When do velocity and acceleration become equal?



UNIT-III

MARKS-14

INTEGRATION

AND

ITS APPLICATION



INTEGRATION

➤ MCQ'S

- 1) $\int x^3 dx = \underline{\hspace{2cm}} + c.$
- 2) $\int x^4 dx = \underline{\hspace{2cm}} + c.$
- 3) $\int x^7 dx = \underline{\hspace{2cm}} + c.$
- 4) $\int \frac{1}{x^2} dx = \underline{\hspace{2cm}} + c.$
- 5) $\int \log x dx = \underline{\hspace{2cm}} + c.$
- 6) $\int \sin x dx = \underline{\hspace{2cm}} + c.$
- 7) $\int \cos x dx = \underline{\hspace{2cm}} + c.$
- 8) $\int (\sin^2 x + \cos^2 x) dx = \underline{\hspace{2cm}} + c.$
- 9) $\int \tan^2 x dx = \underline{\hspace{2cm}} + c.$
- 10) $\int \tan x dx = \underline{\hspace{2cm}} + c.$
- 11) $\int e^{x \log a} dx = \underline{\hspace{2cm}} + c.$
- 12) $\int e^{-\log(\sec x)} dx = \underline{\hspace{2cm}} + c.$
- 13) $\int \frac{1}{1+x^2} dx = \underline{\hspace{2cm}} + c.$
- 14) $\int \frac{1}{\sqrt{a^2-x^2}} dx = \underline{\hspace{2cm}} + c.$
- 15) $\int \frac{1}{x^2+25} dx = \underline{\hspace{2cm}} + c.$
- 16) $\int \frac{1}{\sqrt{1-x^2}} dx = \underline{\hspace{2cm}} + c.$
- 17) $\int \cos(ax+b) dx = \underline{\hspace{2cm}} + c.$
- 18) $\int (\sqrt{1+\sin 2x}) dx = \underline{\hspace{2cm}} + c.$
- 19) $\int_0^1 e^x dx = \underline{\hspace{2cm}} + c.$
- 20) $\int_0^1 x dx = \underline{\hspace{2cm}} + c.$
- 21) $\int_2^5 x^3 dx = \underline{\hspace{2cm}} + c.$
- 22) $\int_{-1}^1 x^3 dx = \underline{\hspace{2cm}} + c.$
- 23) $\int_1^e \frac{dx}{x} = \underline{\hspace{2cm}} + c.$
- 24) $\int_{-1}^1 (x^2 + 1) dx = \underline{\hspace{2cm}} + c.$
- 25) $\int_1^e \log x dx = \underline{\hspace{2cm}} + c.$
- 26) $\int e^x (\tan x + \sec^2 x) dx = \underline{\hspace{2cm}} + c.$
- 27) $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \underline{\hspace{2cm}} + c.$
- 28) Area covered by the curve $x^2 + y^2 = 4$ is _____.



➤ **EXAMPLES (Based on Addition and subtraction)**

- 1) Evaluate $\int 4x^2 + 3x + 9 dx$.
- 2) Evaluate $\int \left(\frac{2x^2 - 3x - 11}{x} \right) dx$.
- 3) Evaluate $\int \left(\frac{3x^2 + 2x - 5}{x} \right) dx$.
- 4) Evaluate $\int \left(\frac{x^3 + 5x^2 + 4x + 1}{x^2} \right) dx$.
- 5) Evaluate $\int \frac{x^2 + 5x + 6}{x^2 + 2x} dx$.
- 6) Evaluate $\int \left(x + \frac{1}{x} \right)^2 dx$.
- 7) Evaluate $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 dx$.
- 8) Evaluate $\int \frac{x^4 + x^2 + 1}{x^2 + 1} dx$.
- 9) Evaluate $\int \frac{x^2 + 4x + 1}{x^3 + x} dx$.
- 10) Evaluate $\int \frac{2 + 3\sin x}{\cos^2 x} dx$.
- 11) Evaluate $\int \frac{4 + 3\cos x}{\sin^2 x} dx$.
- 12) Evaluate $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cdot \cos^2 x} dx$.
- 13) Evaluate $\int \frac{1}{\sin^2 x \cdot \cos^2 x} dx$.
- 14) Evaluate $\int \sec^2 x \cdot \operatorname{cosec}^2 x dx$.
- 15) Evaluate $\int \frac{\cos 2x}{\sin^2 x \cdot \cos^2 x} dx$.
- 16) Evaluate $\int \frac{dx}{1 - \sin x}$.
- 17) Evaluate $\int \frac{\cos x}{1 + \cos x} dx$.

➤ **EXAMPLES (Based on integration by parts)**

- 1) Evaluate $\int x \cdot e^x dx$.
- 2) Evaluate $\int \sin x \cdot x dx$.
- 3) Evaluate $\int x \cdot \cos x dx$.
- 4) Evaluate $\int x \cdot \log x dx$.
- 5) Evaluate $\int x \cdot e^{mx} dx$.
- 6) Evaluate $\int x \cdot e^{3x} dx$.
- 7) Evaluate $\int x^2 \cdot \log x dx$.
- 8) Evaluate $\int x^2 \cdot e^x dx$.
- 9) Evaluate $\int x \cdot \tan^{-1} x dx$.
- 10) Evaluate $\int \log x dx$.



➤ **EXAMPLES (Based on substitution method)**

- 1) Evaluate $\int e^{\sin x} \cdot \cos x \, dx$.
- 2) Evaluate $\int e^{\cos x} \cdot \sin x \, dx$.
- 3) Evaluate $\int e^{\tan x} \cdot \sec^2 x \, dx$.
- 4) Evaluate $\int \sin^5 x \cdot \cos x \, dx$.
- 5) Evaluate $\int \cos x \cdot \sqrt{\sin x} \, dx$.
- 6) Evaluate $\int \frac{\sin(\log x)}{x} \, dx$.
- 7) Evaluate $\int \frac{(\log x)^5}{x} \, dx$.
- 8) Evaluate $\int \frac{2 \tan^{-1} x}{1+x^2} \, dx$.
- 9) Evaluate $\int \frac{e^x(x+1)}{\sin^2(xe^x)} \, dx$.
- 10) Evaluate $\int \frac{e^x(x+1)}{\cos^2(xe^x)} \, dx$.
- 11) Evaluate $\int 2x \cdot e^{x^2} \, dx$.
- 12) Evaluate $\int \frac{x^2}{1+x^6} \, dx$.
- 13) Evaluate $\int 2x \cdot (x^2 + 8)^8 \, dx$.
- 14) Evaluate $\int (2x + 1) \sqrt{x^2 + x + 9} \, dx$.

➤ **EXAMPLES (Based on partial fraction)**

- 1) Evaluate $\int \frac{dx}{x^2 - 3x + 2}$.
- 2) Evaluate $\int \frac{x}{(x+1)(x+2)} \, dx$.
- 3) Evaluate $\int \frac{x+3}{(x-1)(x-2)} \, dx$.
- 4) Evaluate $\int \frac{2x+1}{(x+1)(x-3)} \, dx$.
- 5) Evaluate $\int \frac{2x+3}{(x-1)(x+2)} \, dx$.

➤ **EXAMPLES (Based on definite integration)**

- 1) Evaluate $\int_1^3 (x^2 + x + 1) \, dx$.
- 2) Evaluate $\int_1^2 (x^2 + 4x + 1) \, dx$.
- 3) Evaluate $\int_1^3 (2x^2 + 5x + 1) \, dx$.
- 4) Evaluate $\int_1^2 \frac{x^3 - 1}{x - 1} \, dx$.
- 5) Evaluate $\int_{-1}^1 \frac{x^3 - 8}{x - 2} \, dx$.
- 6) Evaluate $\int_0^1 \frac{2}{1+x^2} \, dx$.
- 7) Evaluate $\int_1^3 \frac{2x}{1+x^2} \, dx$.
- 8) Evaluate $\int_0^2 \frac{x^2}{1+x^3} \, dx$.
- 9) Evaluate $\int_{-1}^2 \frac{x}{x^2 + 3} \, dx$.



10) Evaluate $\int_1^e \frac{(\log x)^2}{x} dx$.

11) Evaluate $\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$.

12) Evaluate $\int_0^1 \frac{x}{x+1} dx$.

13) Evaluate $\int_{-4}^{-3} \frac{x}{7+x} dx$.

➤ **EXAMPLES (Based on definite integration for continuous function)**

1) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$.

2) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\tan x}{\tan x + \cot x} dx$.

3) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sec x}{\sec x + \operatorname{cosec} x} dx$.

4) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx$.

5) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sec x}}{\sqrt{\sec x} + \sqrt{\operatorname{cosec} x}} dx$.

6) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\sin x}}{\sqrt[3]{\sin x} + \sqrt[3]{\cos x}} dx$.

7) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sqrt{\tan x}} dx$.

8) Evaluate $\int_0^{\frac{\pi}{2}} \log(\tan x) dx$.

9) Evaluate $\int_0^{\frac{\pi}{2}} \log(\cot x) dx$.

10) Evaluate $\int_0^5 \frac{\sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}} dx$.

11) Evaluate $\int_0^7 \frac{\sqrt{7-x}}{\sqrt{x} + \sqrt{7-x}} dx$.

➤ **EXAMPLES (Based on application of integration area and volume)**

1) Find the area bounded by the curve $x^2 + y^2 = a^2$.

2) Show that the area enclosed between the parabola $y = x^2$ and lines $x = 2, x = 3$ and x - axis is $\frac{19}{3}$ sq. unit.

3) Find the area of region bounded by $y = 2x^2, x$ - axis and lines $x = 5$.

4) Find the area of region bounded by $y = 3x^2, x = 2, x = 3$ and x - axis.

5) Find the area of region bounded by $y = 4x^2, x = 1, x = 2$.

6) Find the area of region bounded by the curves $y = x^2$ and $y = x$.

7) Find the area of region bounded by the curves $y = x^2$ and the line $y = x + 2$.

8) Find the area of region bounded by the curves $y^2 = 4x$ and $x = 2$.

9) Find the area of region bounded by the curves $y = x^2 - 7x + 10$ and x - axis.

10) Find the area bounded by the curves $x + y = 1$ and the axes.

11) Find the volume of a sphere of radius r by method of integration.



UNIT-IV

MARKS-12

DIFFERENTIAL

EQUATIONS



DIFFERENTIAL EQUATIONS

➤ **MCQ'S**

- 1) The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + 3\left(\frac{dy}{dx}\right)^2 - 5y = 0$ are _____ respectively.
- 2) The order of the differential equation $\left(\frac{d^3y}{dx^3}\right)^2 + 3\left(\frac{d^2y}{dx^2}\right)^4 + x \sin y = 0$ is _____.
- 3) Order of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = xy$ is given by _____.
- 4) The order of the differential equation $x \frac{d^2y}{dx^2} - 5\left(\frac{dy}{dx}\right)^3 - 2y = 14$ is _____.
- 5) The degree of the differential equation $x^2 \frac{dy}{dx} + \sin\left(\frac{d^2y}{dx^2}\right) = 0$ is _____.
- 6) Order of $\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6 = 0$ is _____.
- 7) The order of the differential equation $\frac{d^2y}{dx^2} = \left(3 + \frac{dy}{dx}\right)^3$ is _____.
- 8) The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + 3\left(\frac{dy}{dx}\right)^2 - 9y = 0$ are _____ and _____ respectively.
- 9) For the differential equation $\frac{dy}{dx} + Py = Q$, Integrating factor is _____.
- 10) Integrating factor of $\frac{dy}{dx} + \frac{2y}{x} = e^x$ is _____.
- 11) The integrating factor of the equation $\frac{dy}{dx} = y \tan x + e^x$ is _____.
- 12) The integrating factor of the equation $\frac{dy}{dx} + y = 3x$ is _____.
- 13) The integrating factor of the equation $\frac{dy}{dx} + \frac{y}{x} = x$ is _____.
- 14) The integrating factor of the equation $\frac{dy}{dx} + 2y = e^x$ is _____.
- 15) The integrating factor of the equation $\frac{dy}{dx} + y \tan x = \cos x$ is _____.



➤ **EXAMPLES (Based on Separable variable)**

- 1) Find the order and degree of $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \rho \left(\frac{d^2y}{dx^2}\right)^2$.
- 2) Find the differential equation for $y = a\sin(x + b)$, where a and b are arbitrary constants.
- 3) Form the differential equation whose general solution is $y = A\cos x + B\sin x$.
- 4) Solve $\frac{dy}{dx} = \frac{y}{x}$.
- 5) Solve the differential equation $x \cdot dy + y \cdot dx = 0$.
- 6) Solve $x(1 + y^2)dx - y(1 + x^2)dy = 0$.
- 7) Solve $(1 + x^2)dx = (1 + y^2)dy$.
- 8) Solve $(1 + x^2)dy - (1 + y^2)dx = 0$.
- 9) Solve $\frac{dy}{dx} + x^2 \cdot e^{-y} = 0$.
- 10) Solve $x \frac{dy}{dx} + \cot y = 0$.
- 11) Solve the differential equation $\tan y \, dx + \tan x \cdot \sec^2 y \, dy = 0$.
- 12) Solve the differential equation $\sec^2 x \cdot \tan y \, dx + \sec^2 y \cdot \tan x \, dy = 0$.
- 13) Solve $\sec^2 x \cdot \tan y \, dx + \sec^2 y \cdot \tan x \, dy = 0$ where $y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$.
- 14) Solve $x \cdot \cos^2 y \, dx = y \cdot \cos^2 x \, dy$.

➤ **EXAMPLES (Based on Linear differential equation)**

- 1) Solve $\frac{dy}{dx} + \frac{2y}{x} = \sin x$.
- 2) Solve $\frac{dy}{dx} + 2y = e^x$.
- 3) Solve $\frac{dy}{dx} + y \tan x = \cos x$.
- 4) Solve $\frac{dy}{dx} + y \tan x = \sec x$.
- 5) Solve $\frac{dy}{dx} + y \cot x = \cos x$.
- 6) Solve $\frac{dy}{dx} + y \tan x = \sec^2 x$.
- 7) Solve $x \frac{dy}{dx} - y = x^2$.
- 8) Solve $\cos x \frac{dy}{dx} + y = \sin x$.
- 9) Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$.
- 10) Solve $x \log x \frac{dy}{dx} + y = \log x^2$.
- 11) Solve $(1 + x^2) \frac{dy}{dx} + y = \tan^{-1} x$.
- 12) Solve $x \frac{dy}{dx} + 2y = \log x$.
- 13) Solve $\frac{dy}{dx} + \frac{4x}{x^2+1} y = \frac{1}{(x^2+1)^3}$.



UNIT-V

MARKS-12

STATISTICS



STATISTICS

➤ MCQ'S

- 1) If $x_1, x_2, x_3, \dots, x_n$ be n observations of the ungrouped data then mean of the observation $\bar{x} = \underline{\hspace{2cm}}$.
- 2) The mean of observations 19,15,28,34,27,21,14,24,6,2 is $\underline{\hspace{2cm}}$.
- 3) The mean of the numbers 39, 23, 58,47,50,16 and 61 is $\underline{\hspace{2cm}}$.
- 4) For the information data 32,26,41,35,28,42,36,40,33,42 mean of observation is $\underline{\hspace{2cm}}$.
- 5) Mean of 3, 5,2,6,5 is $\underline{\hspace{2cm}}$.
- 6) The mean value of first eight natural numbers is $\underline{\hspace{2cm}}$.
- 7) The mean of first ten natural numbers is $\underline{\hspace{2cm}}$.
- 8) Mean of first five natural numbers is $\underline{\hspace{2cm}}$.
- 9) Mean value of first five prime numbers is $\underline{\hspace{2cm}}$.
- 10) If mean of 4, 7, 6, K, 5, 9 is 6 then $K = \underline{\hspace{2cm}}$.
- 11) If the mean of observations 15, 7, 6, a, 3 is 7 then a $\underline{\hspace{2cm}}$.
- 12) Mean of 10 observations is 15. Remove one observation 24 from that then find mean of remaining observations.
- 13) The mean height of 30 students is 5.3ft. One reading was entered wrong as 5.2ft. instead of 4.9ft. Find the correct mean.
- 14) The range of the data 17, 15,25,34,32 is $\underline{\hspace{2cm}}$.
- 15) Mean deviation of n observation $x_1, x_2, x_3, \dots, x_n$ using mean $\bar{x}$ is $\underline{\hspace{2cm}}$.



➤ **EXAMPLES (Based on Mean)**

1) Find mean of 23, 20, 21, 17, 20, 23, 18, 19.

2) Find mean from following data.

x	92	93	97	98	102	104	109
f	3	2	3	2	6	3	3

3) Find mean from following data.

x	92	93	97	98	102	104
f	3	2	2	3	6	4

4) Find mean from following data.

Class	15-20	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
fi	3	8	10	19	25	21	6	5	3

5) The freq. distribution of the age of 60 staff members of a college is as below. Find the mean.

Age(year)	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59
No. of staff	5	7	9	11	10	8	6	4

6) Find the mean for the freq. distribution given below.

Class	0-50	50-100	100-150	150-200	200-250	250-300	300-350
fi	10	15	30	20	15	8	2

7) Find the mean of the following grouped freq. distribution.

Marks obtained	21-25	26-30	31-35	36-40	41-45	46-50
No. of students	8	10	24	30	12	16

8) If the mean of the following frequency distribution is 26 then find the missing frequency.

Class	0-9	10-19	20-29	30-39	40-49	Total
Freq.	6	a	17	b	8	60

➤ **EXAMPLES (Based on Mean deviation)**

1) Find the mean deviation from mean for given data 4, 6, 2, 4, 5, 4, 4, 5, 3, 4.

2) Find the mean deviation from mean for following data.

58, 52, 45, 48, 57

3) Find the mean deviation from mean for following data.

168, 164, 172, 169, 178, 173, 173

4) Find the mean deviation from mean for following data.

xi	4	8	11	17	20	24	32
fi	3	5	9	5	4	3	1

5) Find the mean deviation from mean for following data.

xi	3	9	17	23	27
fi	8	10	12	9	5

6) Find the mean deviation from mean for following data.

Class	0-10	10-20	20-30	30-40	40-50
fi	5	8	15	16	6

7) Find the mean deviation from mean for following data.

Class	4-8	8-12	12-16	16-20	20-24	24-28	28-32	32-36	36-40
fi	5	8	18	25	15	12	10	5	2



➤ **EXAMPLES (Based on Standard deviation (S.D.))**

- 1) Calculate the standard deviation for the following data.

6, 7, 10, 12, 13, 4, 8, 12

- 2) Find the standard deviation for the following data.

120, 132, 148, 136, 142, 140, 165, 153

- 3) Calculate the standard deviation for the following data.

69, 67, 68, 66, 69, 64, 63, 65, 72

- 4) Calculate the standard deviation using the following data.

10, 15, 17, 19, 19, 21, 23, 25, 26, 30

- 5) Compute S.D. of the following data.

12, 16, 19, 20, 23, 25, 28, 30, 32, 35

- 6) Find S.D. from the following data.

xi	4	8	11	17	20	24	32
fi	3	5	9	5	4	3	1

- 7) Find S.D. from the following data.

xi	2	4	6	8	10	12	14	16
fi	4	4	5	15	8	5	4	5

- 8) Find the S.D. of the following grouped frequency distribution.

Class	0-10	10-20	20-30	30-40	40-50
frequency	3	8	15	16	6

- 9) Find S.D. for the following data.

Class	0-20	20-40	40-60	60-80	80-100
frequency	12	38	42	23	5

